

The empirical recurrence for OEIS A238149, recovered from its tail

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Abstract

OEIS A238149 records “Empirical recurrence of order 56” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1608$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{56} = -544008960 \neq 0$, so no further tail satisfies anything shorter; any order-56 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A238149 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the upper median of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1608, 15800, 165780, 1792226, 18918620, 202875630, 2177434900, 23302931502, ...

The entry states:

Empirical recurrence of order 56 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:47:15, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1608$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1608$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 56: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 56.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3216$ exact terms from $n = 3$ on, it returns order 56 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{56} c_i a(n-i),$$

c_1	0	c_{15}	−2843829776	c_{29}	4402358503812	c_{43}	−177432324152
c_2	0	c_{16}	−9694985397	c_{30}	−9182949092316	c_{44}	−1582800590936
c_3	268	c_{17}	−8043882910	c_{31}	−9298358136336	c_{45}	332941272
c_4	8637	c_{18}	−30361858428	c_{32}	7394305507578	c_{46}	576677488876
c_5	13326	c_{19}	150120878734	c_{33}	−7374336997072	c_{47}	63727107816
c_6	52534	c_{20}	33227265706	c_{34}	13845364922545	c_{48}	−35796960644
c_7	107110	c_{21}	222250175088	c_{35}	5825135471710	c_{49}	−75076128192
c_8	−2933633	c_{22}	−140106611629	c_{36}	−10453740014428	c_{50}	−60020231088
c_9	−1304172	c_{23}	−1551131414518	c_{37}	3503750065728	c_{51}	−7145493120
c_{10}	−18307757	c_{24}	294878046185	c_{38}	−7949307672871	c_{52}	16208352432
c_{11}	−44580768	c_{25}	−1291325011034	c_{39}	−2149020794826	c_{53}	−7770816
c_{12}	357580094	c_{26}	2611761988677	c_{40}	7097916871835	c_{54}	−2955642048
c_{13}	65116306	c_{27}	5925116227610	c_{41}	508938201176	c_{55}	−227577600
c_{14}	1858939795	c_{28}	−2708752879157	c_{42}	−413163810932	c_{56}	−544008960

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A238149 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{56} - c_1 x^{55} - \dots - c_{56}$. Its constant term is $-c_{56} = 544008960$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 56 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 56, so $\deg q = 56$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 56, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 56, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -544008960 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-56 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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